Explain the need for dimensionality reduction and describe its advantages and limitations.
Differentiate between PCA and LDA.

Explain the role of eigenvalues and eigenvectors in PCA.

Explain the curse of dimensionality. How does it affect machine learning models?
Explain Principal Component Analysis with a suitable example.

Describe Linear Discriminant Analysis with a suitable example.
Explain Canonical Correlation Analysis in detail.

Differentiate between PCA, LDA and CCA.

Compare biological neural network with an artificial neural network.

Construct a diagram of a perceptron and explain its structure.

Construct a diagram of a multilayer perceptron and explain its structure.

Discuss the role of weights in a neural network with a suitable example.

Discuss the advantages and disadvantages of artificial neural networks.

Differentiate between linear and nonlinear activation functions.

Differentiate between the binary step function and the sigmoid function.

Explain linear activation functions and their limitations.
Explain artificial neural network with its application areas.

Explain the architecture of artificial neural network.

Discuss the backpropagation algorithm in detail.(weight update formula)

Explain the architecture of a feedforward neural network.

Compare the structure of a perceptron with a brain neuron with a neat diagram.

Discuss the learning process in multilayer perceptron.

Explain any three nonlinear activation functions in a neural network.

Discuss the universal approximation theorem.